Machine Learning-Based Ranking of Factors Influencing Human Movement Purposes for Supporting Human-Infrastructure Interaction Modeling

Lan Zhang, M.Eng.[1]; and Kaijian Liu, Ph.D., A.M.ASCE[2]

[1]Graduate Student, Dept. of Civil, Environmental and Ocean Engineering, Stevens Institute of Technology, Hoboken, NJ. Email: lzhang29@stevens.edu
[2]Assistant Professor, Dept. of Civil, Environmental and Ocean Engineering, Stevens Institute of Technology, Hoboken, NJ (corresponding author). Email: Kaijian.Liu@stevens.edu

ABSTRACT

Modeling, predicting, and controlling the interactions between humans and civil infrastructure systems can simultaneously improve the operational efficiency of infrastructure systems and the satisfaction of infrastructure users. The first step towards achieving this goal is to model human-to-infrastructure interaction, which in most cases is driven by human movements (e.g., moving from an origin location to a destination requires using the transportation infrastructure connecting the two). To this end, this paper aims to conduct a machine learning-based data-driven analysis to rank the importance of factors influencing human movement purposes, thereby identifying highly-influential factors to support subsequent modeling of human-to-infrastructure interaction. The research methodology included: (1) representing movement instances using spatial and land use, temporal, and demographic features, and (2) conducting feature ranking per movement purpose using the logistic regression algorithm. As a preliminary work, this paper focuses on presenting the research methodology, and analyzing and discussing the feature ranking results.

INTRODUCTION

The rapid increase of the urban population in recent years poses unprecedented challenges to civil infrastructure systems in urban areas. Currently, around 4.4 billion people are living in urban areas, and the number of urban populations is going to be doubled by 2050 (World Bank 2022). However, the capacity of civil infrastructure systems in urban areas cannot keep up with increased demands from the growing population. For example, according to the 2021 Infrastructure Report Card, as the nation's public roadways are serving more users each year, congestion increased at an annual rate of 3% – with 30% of trips on the roadways impacted by severe or extreme congestion – causing people in urban areas to spend an extra of 8.8 billion hours on road and 3.3 billion gallons of fuel each year (ASCE 2021). Similarly, the wastewater infrastructure systems nowadays are already operating at eighty percent of the designed capacity on average (ASCE 2021), hardly capable of accommodating the doubled populations in the near future. Although expanding the capacity of civil infrastructure systems (e.g., building new roads and sewers) appears to be a plausible solution to cope with the challenges, infrastructure capacity expansion is much limited by the shortage of available space in urban areas. Vacant land is rare in urban areas (Haughwout et al. 2008; Kim et al. 2018): much of the land has already been exploited and occupied by existing infrastructure and other physical assets (e.g., residential and commercial buildings). Under such conditions, there is therefore a need to improve the operational efficiency of infrastructure systems to leverage the current given capacity of infrastructure systems to, effectively, provide services to ever-increasing populations in urban areas.

The modeling, prediction, and controlling of the interactions between humans and civil infrastructure systems hold great promise to improve the operational efficiency of infrastructure systems without the need to expand their capacity, while simultaneously enhancing the satisfaction of the needs of infrastructure users. *Human-infrastructure interaction*, as proposed in this study, captures (1) how humans use civil infrastructure systems to satisfy their needs, e.g., capturing the probabilistic distributions of how various population groups use a certain type of road paths to commute to work across time; and (2) how infrastructure systems, such as their topological and functional properties, affect the use patterns of humans, e.g., how the connectivity of transportation infrastructure systems in a community shapes and affects the travel behaviors of humans. The first step towards modeling, prediction, and controlling is to understand and model human-to-infrastructure interactions, i.e., infrastructure use patterns of different population groups. In most cases, infrastructure uses are linked to human movements in urban areas. For example, millions of urban inhabitants move between their home locations and workplaces on a daily basis, increasing the use of transportation infrastructure systems during rush hours, energy infrastructure systems at workplaces during working hours, and wastewater infrastructure systems at home at rest time.

As such, this paper aims to conduct a machine-learning-based data-driven analysis to rank the importance of factors influencing various types of human movements (e.g., work, school, shop, etc.). The ranking allows for identifying highly-influential factors per movement purpose to facilitate the understanding of human movements in urban areas and the selection and weighting of features/factors for movement prediction and generation, all contributing to subsequent human-to-infrastructure interaction modeling. The research methodology used in this study included two main components. First, each human movement instance is described using a set of features representing different influencing factors: (1) the spatial and land use characteristics of travelers' home zones as well the origin and destination locations of the trip, (2) the temporal characteristics of the trip, and (3) the demographic characteristics of traveler(s). Second, the logistic regression algorithm with *L-2* regularization is used to learn from all the movement instances collected in this study to model and predict the purpose of each movement instance. The weights of the resulting logistic regression model for movement purpose prediction are used to rank the importance of the influencing factors for each movement purpose type. As a preliminary work, this paper focuses on presenting the research methodology, and analyzing and discussing the feature ranking results.

BACKGROUND AND KNOWLEDGE GAPS

Ranking the importance of factors affecting human movement in urban areas is a critical step to understand and generate infrastructure use patterns by humans (as part of human-to-infrastructure modeling). In recent years, the availability of travel survey data from different sources offers a great promise to human movement purpose modeling, including the ranking of the factors affecting various purposes, such as the National Household Travel Survey (FHWA 2023), My Daily Travel survey (CMAP 2023), and Citywide Mobility Survey (NYC DOT 2023). Such travel surveys provide detailed information about factors that influence human movement, including spatial-temporal patterns of human movements, socioeconomic characteristics of travelers, and urban land use characteristics. A number of studies have attempted to leverage these travel survey data for the modeling of human movement purpose, with a focus on learning from travel survey data to predict the types of movement purposes. Existing studies for human movement purpose modeling and prediction can be classified into three categories (Gong et al. 2014): (1) rule-based methods (e.g., Stopher et al. 2008; Pereira et al. 2014), which use manually-developed heuristic rules to associate socioeconomic characteristics of travelers and spatial features of trips to different human movement purposes for the prediction; (2) statistical methods (e.g., Chen et al. 2010; Oliveira et al. 2014), which use statistic models such as logistic models to calculate the probability of each human movement purpose based on data about socioeconomic characteristics and spatial features; and (3) machine learning-based methods (e.g., Wu et al. 2011; Xiao et al. 2016), which leverage machine learning algorithms such as artificial neural networks to learn from travel survey data to model and predict human movement purposes.

Despite their importance, existing trip purpose modeling studies are limited in ranking the importance of various factors affecting movement purposes. Accordingly, two knowledge gaps were identified. First, there is a lack of studies that consider the diverse types of movement purposes that exist. For example, existing studies (e.g., Wu et al. 2011; Xiao et al. 2016) either focus on general high-level movements (e.g., indoor versus outdoor) or a few specific movement purposes (e.g., home, work, and school), without offering a fuller picture of human movement purposes (e.g., home, work, work-related, school, escort, shop, meal, social and recreation, errand, and change mode). Second, and more importantly, there is a lack of studies that rank the importance of influencing factors in determining each human movement purpose to explore the relationship between influencing factors and human movement purpose. As discussed above, existing studies mainly focused on predicting movement purposes, instead of ranking the importance of influencing factors. The ranking of the influencing factors allows for identifying high-influential factors that can be used to model human-to-infrastructure interaction, which cannot be achieved by existing purpose prediction models.

RESEARCH METHODOLOGY

To address the aforementioned knowledge gaps, this paper aims to conduct a machine learning-based data-driven analysis to rank the importance of influencing factors in driving different human movement purposes. The research methodology included two components: human movement representation, and logistic regression-based ranking.

Human Movement Representation. Human movement representation aims to represent each movement instance using a representative set of features that can capture the factors influencing various movement purposes. To support the representation, human movement data were collected from the City Mobility Survey (NYC DOT 2023). The City Mobility Survey is an annual human sensing effort conducted by the New York City Department of Transportation (NYC DOT), which covers over 3,000 residents in New York City (NYC) each year to offer valuable sources of information about travel behaviors, preferences, and attitudes of various population groups (NYC DOT 2023). The collected dataset includes a total of 71,620 trips, occurred in NYC between May 22, 2019 through June 30, 2019. Using the movement data collected from the City Mobility Survey, each movement instance is represented using three main types of features capturing the movement purpose-related factors, including the features about (1) the spatial and land use characteristics of travelers' home zones as well the origin and destination locations of the trip, (2) the temporal characteristics of the trip, and (3) the demographic characteristics of traveler(s). Table 1 summarizes the features/factors used to represent each movement instance.

As noted in Table 1, the features are of mixed types, including categorical and numerical features (e.g., travel mode type versus trip duration in minutes). Machine learning algorithms for feature ranking, such as the logistic regression algorithms, are not able to deal with mixed-type features. In addition, the mixing of numerical features into categorical features often makes the resulting machine learning model sensitive to minor fluctuations in numerical feature values, undermining the performance of the machine learning model and the effectiveness of the model in ranking the features. Therefore, two preprocessing steps were conducted to transform the mixed-type features into single-type, one-hot encoded features. First, feature discretization was conducted to discretize the numerical features into discrete intervals. The equal-width binning method, which is one of the most commonly-used discretization methods, was used for the discretization. Each of the numerical features was discretized into ten intervals, with the boundary of each interval defined based on the minimum and maximin values of the feature. Second, the one-hot encoding method was used to numerically represent the categorical features, both originally categorical features and discretized numerical features, numerically for subsequent machine learning.

Table 1. Features/factors used for Human Movement Representation [1].

No.	Feature Type	Feature Name/Description
F01	Spatial and land use	Home citywide mobility survey zone
F02		Origin citywide mobility survey zone
F03		Destination citywide mobility survey zone
F04		*Trip distance (miles)*
F05		Location type for trip origin (e.g., home, work, or school)
F06		Location type for trip destination (e.g., home, work, or school)
F07	Temporal	Travel date (day of the week)
F08		*Trip duration (minutes)*
F09	Demographic	*Number of people in travel party*
F10		Mode type (e.g., for-hire vehicles, commuter rail, or bus)
F11		Household's total annual income: detailed categories [2]
F12		*Household size*
F13		*Number of adults in household*
F14		*Number of children in household*
F15		*Number of workers in household*
F16		*Number of students in household*
F17		*Number of trips*
F18		*Number of vehicles in household*

[1] Features in italic font are numerical features, and features without italic font are categorical features.
[2] Household's total annual income is a categorical feature. The City Mobility Survey uses the following ten categories to describe total annual income for each household: Under \$15,000; \$15,000 to \$24,999; \$25,000 to \$34,999; \$35,000 to \$49,999; \$50,000 to \$74,999; \$75,000 to \$99,999; \$100,000 to \$149,999, \$150,000 to \$199,999; \$200,000 to \$299,999; and \$300,000 or more.

Logistic Regression-Based Ranking. A set of logistic regression models were trained by learning from the collected movement data to predict the purpose for each movement instance, thereby assigning a score to each input feature to assess the importance of the features in predicting and explaining the target movement purpose at hand. The logistic regression algorithm with *L-2* regularization (i.e., the ridge regression) was chosen and used for the ranking in this study, because it is commonly used for data-driven feature selection and ranking. Specifically, the logistic regression algorithm aims to minimize the cost function defined in Eq. (1) for feature ranking (Zakharov and Dupont 2011; Conroy and Sajda 2012), where $P(X_i) = 1/1+exp(-X_i * W)$ and is the probability of movement instance X_i being classified as a positive example of a movement purpose, y_i is the gold-standard class of X_i and could be $+1$ for positive examples or -1 for negative examples, W is a vector of weights for the input features, and n is the number of movement instances/training examples in the created dataset (n = 71,620 in this study). To minimize the cost function, the training of a logistic regression needs to learn the weights for the input features, such that the learned weights (1) can be used to accurately differentiate the positive and negative examples to reduce the cost associated with the first item in the cost function and (2) are shrunken in magnitude for features not discriminative of the target purpose type – hence not important in influencing the movement purpose – to reduce the cost associated with the second item.

In this study, using the logistic regression algorithm, a total of ten models for ranking the influencing factors/features were developed, with each focusing on ranking the importance of the factors with respect to one of the following ten movement purpose types: "home", "work", "work-related", "school", "escort", "shop", "meal", "social and recreation", "errand and other", and "change mode". The one-versus-all approach was followed for the development of each model: movement instances belonging to the target purpose type were set as the positive examples and the rest of the instances (irrespective of their original purpose types in the dataset) were set as the negative examples – resulting in ten regression models with each corresponding to a purpose type defined above. Prediction accuracy was used to evaluate the performance of the model training, to ensure that each model was properly trained and the resulting importance score for each feature was reliable. In this study, prediction accuracy is the ratio of the number of correctly-predicted movement instances to the total number of movement instances in the dataset. During the training process for each model, the predicting accuracy was monitored to determine whether or not to continue the training iteration. A model was considered to be fully trained if the accuracy did not improve for 15 iterations. For each movement purpose, the weight of each feature in the logarithmic space was then used to assess the importance of each feature for the ranking.

$$\boldsymbol{min} \sum_{i=1}^{n} \left(-y_i \log\left(P(X_i)\right) - \left(1 - y_i\right) \log\left(1 - P(X_i)\right)\right) + \frac{1}{2} W^T W \tag{1}$$

PRELIMINARY RESULTS AND DISCUSSION

Figure 1 shows the accuracy of each logistic regression-based feature ranking model in predicting the target movement purpose type. Overall, the models performed well: on average, they achieved an average accuracy of 92.6% across all ten models. This indicates that the models are reliable and thus can be used to rank the importance of the features. However, the models for predicting movement purpose "shop", "social/recreation" and "change mode" achieved an accuracy of 83.9%, 88.5%, and 86.1%, respectively, which are lower than 90.0%. These lower accuracies may be attributed to two main reasons. First, the number of movement instances of each purpose type varies, with some purpose types having a large number of instances and some having a small number. For example, the collected dataset includes a total of 71,620 movement instances, and only 8,275 trips belong to the movement purpose "social/recreation". As a result, the model for this movement purpose was trained using a highly imbalanced dataset: the ratio of positive examples to negative examples was roughly 1:9. The imbalance limited the algorithm training process, which made the positive examples insufficiently learned and thus led to lower accuracy.

Second, compared to the movement instances of the other purpose types, the destinations and the characteristics of the destinations for movement instances of “shop”, “social/recreation” and “change mode” are more diverse and uncertain. For example, movements related to “home” were mainly destinated to the settled home locations, and movements related “work” were mainly destinated to the settled work locations. However, movements related “shop”, “social/recreation” and “change mode” are associated with diverse types of destinations and fewer regularities, which may have led to the lower accuracies of the three models.

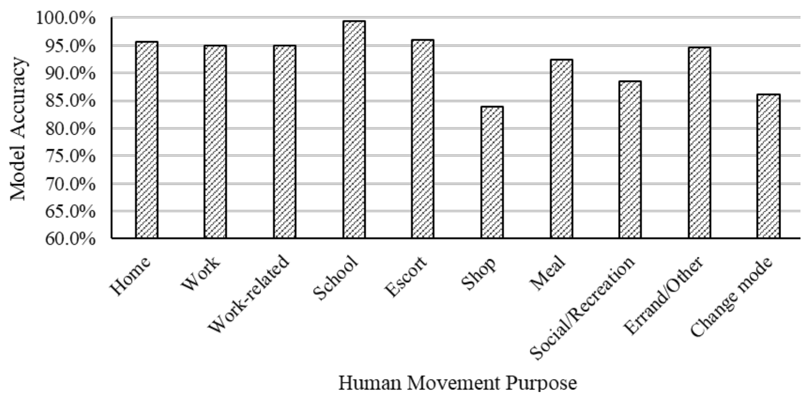

Figure 1. Accuracy of logistic regression-based movement prediction models.

Figure 2 shows the importance scores and the ranking of the features in modeling each movement purpose. Two main conclusions were drawn from the ranking results. First, the importance of the features/factors varied across movement purpose types. For example, for the movement purpose “home”, the top three features were trip distance, trip duration and number of people in travel party (F4, F8 and F9); for the movement purpose “work”, the top important features were household size, number of children in household, and trip duration (F12, F14 and F8); for the movement purpose “social/recreation”, the top important features were number of students in household, trip duration and number of children in household (F16, F8 and F14). While all the three movement purposes included trip duration (F8) as an important feature for the purpose prediction, the other important features for each movement purpose were different. This indicates that when modeling human movements to support human-to-infrastructure interaction modeling, different movement purposes should be modeled separately, with each purpose modeled with respect to the factors important to the modeled purpose type.

Second, despite the variances, some features are common top features across the movement purpose types, and some are not common. The following features are ranked as top features for eight, five, and four purpose types, respectively: trip duration, number of students in household, and trip distance (F8, F16, and F4). The following features, however, are not even ranked as top features for the selected purpose types: home citywide mobility survey zone, origin citywide mobility survey zone, destination citywide mobility zone, location type for trip origin, location type for trip destination and household’s total annual income (F1, F2, F3, F5, F6, and F11). While trip duration was ranked as an important feature for eight movement purpose types across all ten types, some features are not ranked as important features for any movement purpose type. As a result, identifying common and less common features in movement purpose modeling can facilitate the selection of high-influential features for human movement modeling and prediction, to support subsequent human-to-infrastructure interaction modeling.

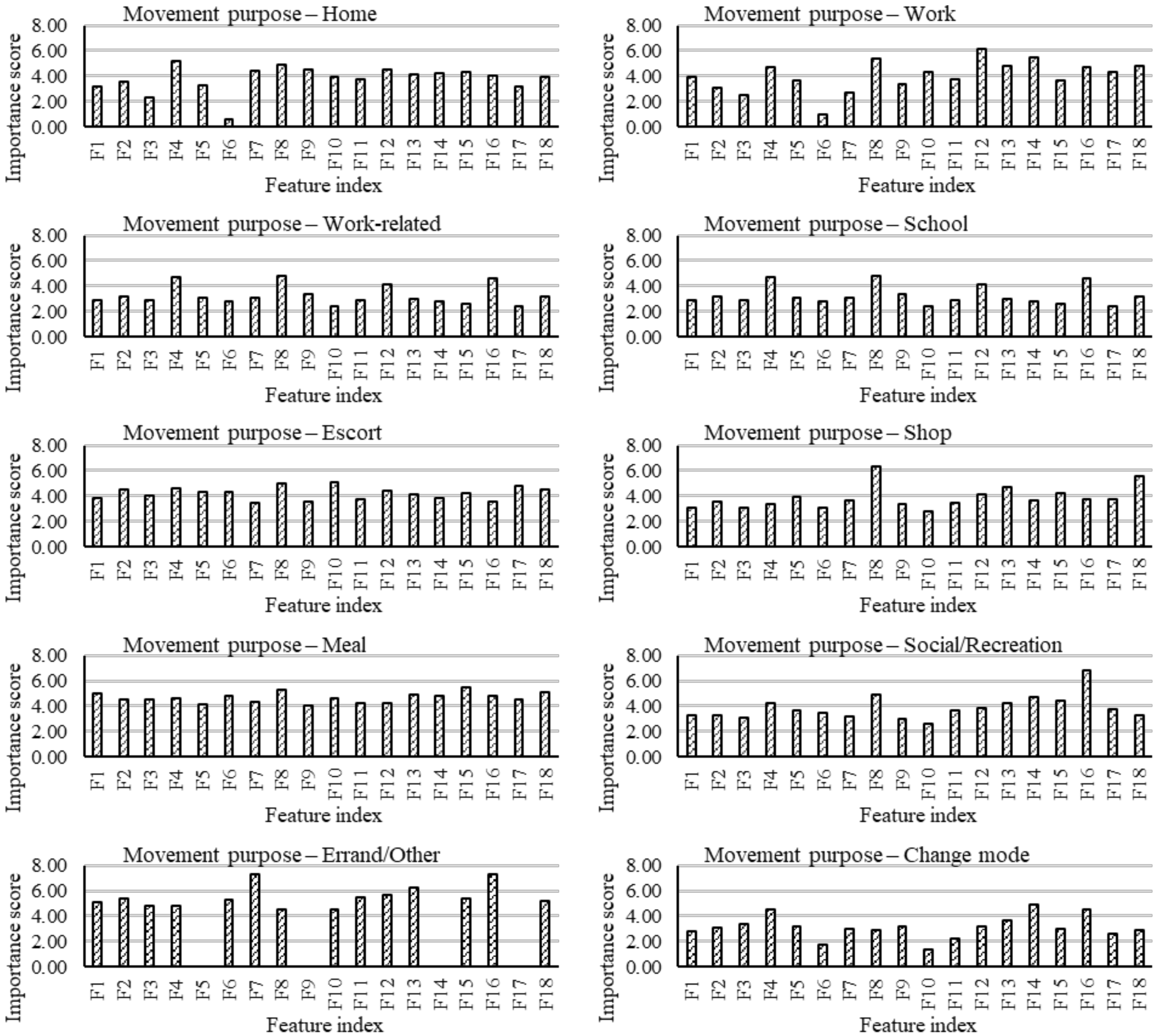

Figure 2. Feature ranking results for human movement purpose modeling.

LIMITATION, CONCLUSION, AND FUTURE WORK

This paper conducted a machine learning-based data-driven analysis to rank the importance of factors affecting human movement purposes. The analysis included two main components: (1) human movement representation using features/factors about spatial and land use characteristics of the origin and destination locations, the temporal characteristics of the trip, and the demographic characteristics of travelers; and (2) logistic regression-based ranking models that use the logistic regression algorithm with *L-2* regularization to adjust the weights of the features for the ranking. Two main conclusions were drawn from the experimental results. First, the importance of the features/factors varied across movement purpose types. Second, despite the variances, some features are common top features across the movement purpose types, and some are not common.

Two limitations of this preliminary study are acknowledged. First, only the City Mobility Survey, which captures the movements of NYC residents, was used for the ranking. As a result, the ranking results may not be generalizable to other cities due to the uniqueness of the city. Second, the ranking was conducted using the logistic regression algorithm, since it is commonly used for feature selection and ranking. However, the prediction performance for some purpose types was lower than 90%. As a result, the feature ranking results from the models with lower performance may not be as reliable as those with high performance. To address these limitations, in their future work, the authors plan to devote their efforts in two directions: (1) enriching the human movement purpose modeling by also modeling influencing factors from other travel survey sources to improve the representativeness of the movements and thus improve the generalizability of the ranking results; (2) conducting deep learning with leave-one-out feature analysis to improve the model prediction as well as feature importance ranking. These efforts could further advance our knowledge of movement purpose modeling to support human-infrastructure interaction modeling, prediction, and control, thereby offering more opportunities to improve the operational efficiency of infrastructure systems and the satisfaction of infrastructure users.

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